

Constant-Wind Zermelo Navigation Beyond the Randers Chamber

Route-A source-local certificate HCS-C305

3 September 2026

Abstract

For every finite $d \geq 1$, constant wind $W \in \mathbb{R}^d$, control cap $c \geq 0$, and target, we give the complete minimum-time atlas for $\dot{x} = W + u$, $|u| \leq c$. A single theorem covers weak, critical, and strong wind, including exact reachability, value formulas, and all attainable times. It also proves the unique constant optimizer, HJB equation, symmetries, Mach-cone regularity, and all zero-data faces. A 744-cell archive and independent checker, symbolic, replay, mutation, and deterministic-build lanes supply regression evidence without replacing proof.

1 The all-chamber value theorem

Consider measurable controls u satisfying $|u(t)| \leq c$ a.e. and

$$\dot{x} = W + u, \quad x(0) = 0. \quad (1)$$

For a target y , write

$$w = |W|, \quad p = W \cdot y, \quad r = |y|, \quad a = w^2 - c^2, \quad D = p^2 - ar^2.$$

Theorem 1 (Complete constant-wind atlas). *At exact time $t \geq 0$, the reachable set is*

$$\mathcal{R}(t) = Wt + ct\overline{B}. \quad (2)$$

For $y \neq 0$, minimum time and all attainable times are as follows.

1. If $w < c$, every y is reachable,

$$T(y) = \frac{\sqrt{p^2 + (c^2 - w^2)r^2} - p}{c^2 - w^2}, \quad \mathcal{I}(y) = [T, \infty). \quad (3)$$

2. If $w = c > 0$, reachability is equivalent to $p > 0$, and then

$$T(y) = \frac{r^2}{2p}, \quad \mathcal{I}(y) = [T, \infty). \quad (4)$$

3. If $w > c$, reachability is equivalent to $p > 0$ and $D \geq 0$. Then

$$T_{\mp}(y) = \frac{p \mp \sqrt{D}}{w^2 - c^2}, \quad T(y) = T_-(y), \quad \mathcal{I}(y) = [T_-, T_+]. \quad (5)$$

For every nonzero reachable target the time-optimal control is unique up to null sets and constant:

$$u_*(s) = y/T(y) - W, \quad |u_*| = c. \quad (6)$$

Proof. Integrating (1) shows that control averages fill $c\overline{B}$, proving (2). Thus y is attainable at t exactly when

$$|y - Wt|^2 \leq c^2 t^2 \iff f(t) = at^2 - 2pt + r^2 \leq 0. \quad (7)$$

For $y \neq 0$, $f(0) > 0$. If $a < 0$, its roots have opposite signs and the positive root is (3); the inequality holds thereafter. If $a = 0$, it is linear and crosses zero at positive time exactly when $p > 0$, giving (4). If $a > 0$, it has a nonnegative-time sublevel interval exactly when $p > 0$ and $D \geq 0$. Both roots are then positive and the sublevel set is precisely (5); in particular the minimum is the smaller root, not T_+ .

At first contact, $|y - WT| = cT$. Any optimal control obeys

$$\left| T^{-1} \int_0^T u(s) ds \right| = c \leq T^{-1} \int_0^T |u(s)| ds \leq c.$$

Equality in the strictly convex Euclidean triangle inequality forces the control to equal $y/T - W$ a.e. For $c = 0$ this conclusion is immediate. $\square$

2 Value geometry, HJB, and exact boundaries

Proposition 1 (Symmetries and HJB). *For every orthogonal Q , $\lambda > 0$, and $s > 0$,*

$$T_{QW,c}(Qy) = T_{W,c}(y), \quad T_{W,c}(\lambda y) = \lambda T_{W,c}(y), \quad T_{sW,sc}(y) = s^{-1}T_{W,c}(y). \quad (8)$$

On each interior component where T is finite and $y \neq 0$, it is smooth and

$$W \cdot \nabla T + c|\nabla T| = 1. \quad (9)$$

Proof. Equation (8) follows from (1), or directly from p, r, w, c . First assume $c > 0$. At first contact put $F(y, T) = |y - WT|^2 - c^2T^2 = 0$. Implicit differentiation gives

$$\nabla T = \frac{y - WT}{p - (w^2 - c^2)T}. \quad (10)$$

The denominator is $\sqrt{D} > 0$ on the selected smooth radical branch and is $p > 0$ in critical wind. Using $|y - WT| = cT$ in (10) yields (9) with the displayed plus sign. If $c = 0$ and $d > 1$, the reachable ray has empty Euclidean interior. If $c = 0$, $d = 1$, and $W \neq 0$, write $e = W/|W|$ and $y = se$ on the open forward ray. Then $T = s/|W|$, $\nabla T = e/|W|$, and $W \cdot \nabla T = 1$ directly. $\square$

The formulas remain exact on their stated faces, but the domains change.

Face	Exact conclusion
$y = 0$	$T = 0$; every $t \geq 0$ is feasible if $w \leq c$, only $t = 0$ if $w > c$.
$W = 0 < c$	$T(y) = y /c$ for every target.
$c = 0, W \neq 0$	only $y = tW$, $t \geq 0$, is reachable; $u = 0$ is forced.
$W = c = 0$	only the zero target is reachable.
$w = c > 0$	nonzero finite-value domain is the open half-space $p > 0$.
$0 < c < w$	full finite-value domain is the closed cone $\{0\} \cup \{y \neq 0 : p > 0, D \geq 0\}$.
$D = 0$	double time root and square-root loss on a nontrivial cone boundary.

In one dimension with $0 < c < w$, the nonzero reachable ray lies in the smooth interior algebraically; a nontrivial angular cone boundary requires higher dimension. When $c = 0$, the ray is handled separately. Weak wind is the gauge of a velocity ball containing the origin, but the strong-wind value is not a global Finsler norm because it is infinite outside its cone.

3 Evidence, hostile audit, and Route-A boundary

The archive has 29 exact cases: zero wind; generic weak wind; reachable and unreachable critical targets; strong-wind axis, interior, double-root cone, exterior, and backward cases; zero cap; zero velocity; and zero targets in all chambers. Twelve smooth probes reconstruct (10), (9), and both scalings. Together with eight boundary rows this is 744 audited leaves. An independent checker performs 734 assertions, a separate SymPy lane checks 27 identities, isolated replay is byte-exact, and 85 repaired-hash or parser mutations must all fail.

The hostile suite changes the selected strong root, finite time window, critical sign, zero-target time set, optimizer, HJB sign and values, exact keys/types/IDs, and Route-A semantics. Duplicate/nonfinite JSON and duplicate/anchor/alias/merge YAML are rejected. Three round variants are rebuilt twice at a fixed epoch; the final PDF is the round-2 alias. Warnings, fonts, text, pages, and rasterization are checked. These finite lanes regress the theorem; the global proof is the quadratic argument (7).

Collision and claim boundaries. C222 is a second-order double-integrator bang–bang system; C270 uses Heisenberg sub-Riemannian geometry; C268 is an uncontrolled constant-field Lorentz flow. C305 closes first-order controlled Euclidean navigation in every finite dimension and all wind chambers. It does not claim variable wind, obstacles, state constraints, manifolds, or literature novelty.

Under NO_BAD_EULER_OR_ROOT_NUMBER, the exact tuple is

$$(A0_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_FAIL).$$

Travel data are not arithmetic local factors (A0); point-to-point controls are not a primitive periodic-orbit ledger (A1); physical time is not a rational-prime clock (A2); no target determinant or functional equation is constructed (A3); HJB supplies no self-adjoint Hilbert–Pólya operator or target-zero match (A4). Therefore the overall verdict is ROUTE_A_REJECTED, and Route B stays locked. No target Euler factor, root number, automorphy, divisor law, functional equation, or zero correspondence is asserted.

Reproducibility and AI use. The release retains canonical evidence, independent executable lanes, three substantive manuscript revisions, and a self-excluding manifest. A generative language model assisted drafting and code scaffolding; independent checks audit formulas, artifacts, and scope, and final responsibility remains with the authors.

Source lineage

Zermelo’s 1931 paper is the historical owner token for the navigation problem. Bao, Robles, and Shen provide navigation/Randers geometric context. The citations establish lineage, not novelty or priority.

References

- [1] E. Zermelo, “Über das Navigationsproblem bei ruhender oder veränderlicher Windverteilung,” *ZAMM* 11 (1931), 114–124. DOI: [10.1002/zamm.19310110205](https://doi.org/10.1002/zamm.19310110205).
- [2] D. Bao, C. Robles, and Z. Shen, “Zermelo navigation on Riemannian manifolds,” *J. Differential Geom.* 66 (2004), 377–435. DOI: [10.4310/jdg/1098137838](https://doi.org/10.4310/jdg/1098137838).